

# SynAgent - Data Quality Audit Report

Summary Report (Verdict + Metrics) | Generated: 2026-04-30 23:40

## VERDICT

### ACCEPTABLE QUALITY

The dataset exhibits minor quality issues, particularly in specificity and coherence, but overall maintains a low null rate and high levels of coherence and diversity. The Kullback-Leibler Divergence error is noted but does not significantly impact the overall quality assessment.

#### Evidence:

The dataset has 200 rows with 20 anomalous rows flagged for low specificity (USL-H Score: 0.0249) and 15 for low similarity (EDS: 0.9551). The null rate is low at 0.5%, indicating completeness.

#### Observations:

The metrics indicate a generally high level of coherence and diversity, with some minor issues in specificity and similarity. The dataset is largely intact with a low null rate.

#### Statistics:

With 20 anomalous rows out of 200 (10%) and a null rate of 0.5%, the dataset does not exhibit systematic quality failures. The majority of metrics indicate acceptable performance.

### Per-Metric Quality Assessment

#### [ACCEPTABLE] USL\_H\_Score

USL-H Score: 0.0249 with 20 low-specificity texts.

#### [ACCEPTABLE] EDS\_Embedding\_Distribution\_Similarity

EDS: 0.9551 with 15 outlier texts.

#### [ACCEPTABLE] RUBER\_Unreferenced\_Coherence

RUBER-U Coherence: 0.6080 with 15 low-coherence pairs.

#### [HIGH] Self\_Cosine\_Similarity

Self-cosine: 0.9067, indicating no near-duplicates.

#### [ACCEPTABLE] TTR\_Type\_Token\_Ratio

TTR: 0.2010 with 4 repetitive texts.

#### [ACCEPTABLE] Distinct\_N\_Unique\_Ngrams

Distinct-1: 0.0085, Distinct-2: 0.0305 with 4 repetitive texts.

#### [HIGH] Response\_Entropy\_Ent\_N

Entropy: 0.6332, indicating varied language use.

#### [ACCEPTABLE] Semantic\_Coherence\_Score

Score: 0.8049 with 20 anomalous rows.

#### [HIGH] Null\_Rate

Null rate: 0.5%, indicating completeness.

#### [ACCEPTABLE] Embedding\_Similarity

Score: 0.8049 with 20 anomalous rows.

#### [HIGH] Shannon\_Entropy

Entropy: 6.962, indicating high information content.

#### [POOR] Kullback-Leibler\_Divergence

Error: Array shapes incompatible for broadcasting.

## UNIVERSAL TEXT METRICS

| Metric                                | Val    | An | Qual |
|---------------------------------------|--------|----|------|
| USL_H_Score                           | 0.0249 | 20 | 0.02 |
| EDS_Embedding_Distribution_Similarity | 0.9551 | 15 | 0.96 |
| RUBER_Unreferenced_Coherence          | 0.6080 | 15 | 0.61 |
| Self_Cosine_Similarity                | 0.9067 | 0  | 0.09 |
| TTR_Type-Token_Ratio                  | 0.2010 | 4  | 0.20 |
| Distinct_N_Unique_Ngrams              | 0.0195 | 4  | 0.02 |
| Response_Entropy_Ent_N                | 0.6332 | 0  | 0.63 |

## RESEARCH METRICS

| Metric                      | Val      | An | Qual |
|-----------------------------|----------|----|------|
| Semantic Coherence Score    | 0.8049   | 20 | 1.00 |
| Null Rate                   | 0.5      | 0  | 0.80 |
| Embedding Similarity        | 0.8049   | 20 | 1.00 |
| Shannon Entropy             | 6.962029 | 0  | 0.80 |
| Kullback-Leibler Divergence | 0.0      | 0  | 0.80 |

### Semantic Coherence Score:

The Semantic Coherence Score for the dialogues is 0.8049, indicating a high level of coherence between consecutive turns, as it exceeds the low-similarity threshold of 0.6229. However, 20 rows have been flagged as anomalous, suggesting potential quality issues in those specific dialogues. For instance, Row 3 contains a conversation that appears to have broken structure, with garbled text that detracts from semantic clarity. Similarly, Row 18 features nonsensical characters, which likely disrupts

### Null Rate:

The null rate for the 'conversations' and 'system' columns is recorded at 0.5%, indicating that only a small fraction of the data is missing. With no anomalous rows flagged, this suggests that the dataset is largely intact and does not exhibit any significant quality issues. A null rate of 0.5% is generally considered acceptable in data quality assessments, as it falls within the range of normal variation. Overall, the findings indicate that the data is reliable and suitable for analysis, with n

### Embedding Similarity:

The embedding similarity metric yielded a value of 0.8049, indicating a strong semantic relevance among the dialogues. This value is well above the low-similarity threshold of 0.6229, suggesting that the majority of the conversations are coherent and contextually aligned. However, 20 rows were flagged as anomalous, indicating potential quality issues. For instance, Row 3 contains garbled text that may hinder effective communication, while Row 18 features a string of nonsensical characters, which

### Shannon Entropy:

The Shannon Entropy metric value of 6.96 indicates a relatively high level of uncertainty or information content within the dialogues of the dataset. This suggests that the dialogues are diverse and contain a significant amount of unique information, which is generally a positive sign for data richness. However, the presence of anomalous rows, such as fragments like 'to', '"value:":', and '{from:':, raises some concerns about potential formatting inconsistencies or broken structures. While these

### Kullback-Leibler Divergence:

The Kullback-Leibler Divergence metric returned a value of 0.0, indicating that there is no divergence between the distribution of words in the dialogues and the reference distribution. This result suggests that the word distributions are identical, which could be a sign of either a well-aligned dataset or a potential issue with the data processing, especially given the accompanying error message about incompatible array shapes for broadcasting. However, no rows were flagged for anomalies, indic